

DILLON DROBENA

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SUMMARY

Full-stack product engineer and startup cofounder with 10+ years of experience spanning real-time defense radar systems and modern web/mobile products shipped end to end. Deep expertise across the TypeScript / Next.js / React Native / tRPC / Prisma stack, with hands-on ownership of architecture, DevOps, and AI/ML (computer vision + LLMs). Open-source author. Previously held U.S. security clearance (inactive).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Go (Golang), Python, C++, C, SQL, MATLAB
Frontend: Next.js, React, React Native (Expo), Tailwind CSS, Material UI
Backend: tRPC, Node.js, NestJS, FastAPI, .NET Core, Express, Prisma, gRPC / Protocol Buffers, REST / OpenAPI
AI / ML: TensorFlow, MediaPipe, OpenCV, scikit-learn, LLM integration (Anthropic)
Data & Infra: PostgreSQL / PostGIS, Redis, Docker, Kubernetes, Azure
Practices: Turborepo monorepos, microservices, CI/CD, real-time systems

PROFESSIONAL EXPERIENCE

Cofounder & Backend Team Lead — Nexrage Studios May 2020 – Present
Dallas, TX

- Cofounded a software studio; partner in company strategy, client bidding, and overall technical direction.
- Architected and shipped a production **identity-verification platform (FaceID-based)** across dev/stage/prod environments using Next.js, tRPC, Prisma, PostgreSQL, Redis, and Azure, with OpenAPI/Swagger-documented APIs.
- Led backend architecture across multiple client products, standardizing on type-safe tRPC APIs, Prisma schemas, and **Turborepo monorepos** sharing code between Next.js web and Expo (React Native) mobile apps.
- Built **computer-vision ML services** for real-time pose estimation and movement classification (TensorFlow, MediaPipe) served via FastAPI, plus an LLM-powered audit tool (Anthropic API) that crawls sites and returns scored, prioritized reports.
- Engineered **Go microservices** over gRPC / Protocol Buffers with REST gateways, backed by PostgreSQL/PostGIS and Redis, and maintained containerized infrastructure with Docker and Kubernetes.
- Authored and published an **open-source npm package** (tRPC logging middleware) used in production services.

Senior Software Engineer I — Raytheon Technologies — Space & Airborne Systems Jan 2016 – May 2020
Dallas, TX

- Engineered software for **airborne multi-mode surveillance and tactical radar systems** in C, C++, and MATLAB.
- Developed real-time software on VxWorks embedded RTOS for mission-critical airborne platforms.
- Served as on-site subject-matter expert for customer integration efforts and cross-disciplinary engineering collaboration.
- Managed build/release processes and software-metrics reporting for multimillion-dollar programs under **CMMI Level 5** standards.
- Held U.S. Government security clearance (now inactive).

SELECTED PROJECTS

- **Identity Verification Platform** — Cross-industry FaceID identity platform; multi-tenant, multi-environment, OpenAPI-documented. *[Next.js, tRPC, Prisma, Azure, Redis]*
- **AI Search Visibility Auditor** — Built an Answer Engine Optimization tool that crawls a site, simulates how LLM answer engines perceive it, and generates a scored report with prioritized, copy-paste fixes. *[Next.js, Anthropic SDK, Stripe]*
- **Computer-Vision ML Service** — Real-time pose estimation and movement classification API trained on a dataset of 100M+ athlete data points; powers companion Next.js + Expo apps. *[TensorFlow, MediaPipe, FastAPI, Docker]*
- **Real-Time Prediction-Market Trading Platform** — Event-driven trading system with WebSocket order matching at sub-millisecond latency, LMSR automated market-making, and concurrent multi-market support; Next.js web + React Native mobile clients. *[Go, Kafka, Redis, PostgreSQL, Kubernetes]*

EDUCATION

M.S., Computer Science University of Texas at Dallas December 2015
B.S., Computer Science University of Texas at Dallas December 2014